Holistic Accuracy and the Lockean Thesis

Abstract. Does maximizing expected accuracy force a Lockean threshold rule for belief even when accuracy is holistic? Dorst argues that it does. I identify a gap in the derivation and give a counterexample on the unrestricted space of belief states. A score that adds a premium for having no false beliefs is strictly truth-directed and makes the rational assessment of final states independent of their order of formation. Yet, for every positive premium, it permits belief in a proposition at one credence and requires withholding it at a higher credence. The optimum combines a selected core hypothesis with a probability threshold. I also characterize a pair of sufficient repairs to the derivation: together they recover separability on states without complementary pairs. This does not establish that separability is necessary for every Lockean representation. It establishes that truth-directed accuracy and order independence alone do not settle the form of the bridge from credence to belief.

1. Introduction

Accuracy-first epistemology holds that rational doxastic states are those that best promote accuracy by the agent's own lights. With a separable score for outright belief, positive for true beliefs and negative for false ones, it yields a Lockean thesis: believe a proposition just in case your credence in it exceeds a threshold. The separable version of this result is due to Easwaran (2016; see Dorst 2019, §3). Dorst (2019) extended it in two directions. Letting the values of true and false belief vary with proposition and context yields variable thresholds (his Corollary 2). And, he argued, dropping Separability, the assumption that a belief state's accuracy is the sum of the accuracies of its members, changes nothing: given a further premise, Sequence-Irrelevance, Lockeanism follows anyway (his Proposition 5). The resulting claim is that the Lockean form survives a substantial relaxation of separability.[1]

The broad robustness claim needs qualification. The counterexample below preserves unrestricted choice, truth-directed value, and order-independent assessment of final states. It nevertheless violates a fixed, proposition-specific threshold rule. Theorem 1 establishes that violation for every positive value of a premium for an error-free record. Theorem 3 characterizes the resulting optima. Theorem 2 diagnoses the derivation by identifying a pair of sufficient additional assumptions that restore a separable score. It does not prove that every successful derivation of Lockeanism must use them.

The thesis does not say that Lockeanism or separability is false. Nor does it say that holistic accuracy guarantees consistency or deductive closure: the characterization below states exactly when those properties hold. The question is what follows from the advertised minimal premises.

The closest prior work makes the novelty claim narrow. Rothschild (2023) develops additive scoring representations of Lockean belief and explicitly invites investigation of weaker additivity assumptions. Wang (2025, §§2 and 6) studies general expected-utility rules on deductively cogent belief states; his framework permits nonadditive utility and derives convexity constraints. Non-Lockean expected-utility belief is therefore already established. The contribution is an unrestricted-option counterexample directed at Dorst's §4.2, together with a characterization of one holistic score. It does not obtain its result by excluding inconsistent or nonclosed states. Holguín (2022, §5) independently argues that rational thinking need not be monotone in credence. Here that behavior is obtained within an accuracy-maximizing model.

[1] References to Dorst are to the published article, especially pp. 192–195 and 206–207.

The stakes extend beyond formal epistemology. If accuracy alone settled the bridge between credence and belief, the Lockean would win that debate by default. If, as I argue, the shape of the bridge depends on how we value combinations of true beliefs, then the debate over Lockean, closed and guess-like belief becomes a debate about epistemic value, and accuracy-first epistemology is a framework for conducting it rather than a verdict.

2. Dorst's derivation

Let W be a finite set of worlds, let propositions be subsets of W, and let a belief state B be any set of propositions. Accuracy First leaves inconsistent, nonclosed and empty belief states among the alternatives (Dorst 2019, §3). An accuracy measure assigns a real number $A(B, w)$ to each state at each world. Given a probability function C on W, the expected accuracy is $EA_C(B) = \sum_w C(w)A(B, w)$, and Maximize Expected Accuracy says that B is rational only if no state has higher expected accuracy.

Under Separability with proposition-dependent values, $A(B, w) = \sum_{p\in B} v_p(w)$, where $v_p(w) = T_p > 0$ if $w \in p$ and $v_p(w) = F_p < 0$ otherwise. Then $EA_C(B) = \sum_{p\in B} e_p$, where

$$e_p = C(p)T_p + (1 - C(p))F_p.$$

Every optimum includes p when $e_p > 0$ and excludes it when $e_p < 0$; inclusion is optional at equality. The threshold is $\tau_p = -F_p/(T_p - F_p)$ (Corollary 2). Dorst also imposes Conservativeness: in the uniform case $|T| < |F|$, which makes $\tau > 1/2$.

Dorst's §4.2 allows the value of a whole belief state to depend on its other beliefs. He considers adding a proposition to a background state, first when it is true and then when it is false. The argument assigns a positive true-belief increment and a negative false-belief increment, uses them to derive a threshold for the final addition in a sequence, and invokes order independence to extend the threshold to belief generally (2019, pp. 192–195, 206–207).

There are two distinct issues here. A true addition can improve a score at every world where it is true without improving it by the same amount at all those worlds. And invariance under rearranging a sequence does not by itself make the value of an addition independent of which other propositions the final state contains. Section 4 separates these claims.

I read Sequence-Irrelevance as a constraint on rational assessments: holding the other factors fixed, changing the order in which a final state was formed does not alter its rational status. A single objective over whole states satisfies that constraint. It need not make greedy, locally optimal additions lead to a globally optimal state in every order. Requiring that stronger property would constrain the optimization procedure as well as the rationality of its outcome. The counterexample addresses the former, stated order-independence principle; a defender may instead adopt the stronger requirement, but must defend it as an additional restriction on holism.

3. The counterexample

Define the record measure with premium $\beta > 0$:

$$A_\beta(B, w) = \sum_{p\in B} v_p(w) + \beta \cdot [\, w \in \bigcap B \,],$$

where $\bigcap \emptyset = W$ and $[\cdot]$ is 1 if the bracketed condition holds and 0 otherwise. A belief state earns the separable score of its members, plus β at exactly those worlds where everything it contains is true. The premium is the value of an error-free record. One rationale is inferential: a belief state whose members are all true is one from which any amount of multi-premise deduction yields only truths. A body of belief can thus be valuable as a whole, which is the holist's thought. Dorst motivates separability by an exam scored question by question (2019, §3). Many exams also reward a perfect paper, and the record measure scores belief that way. The measure is purely alethic: $A_\beta(B, w)$ depends only on B and on which of its members are true at w.

It meets the stated conditions of Dorst's §4.2 on the final-state reading just specified. Adding a true p raises A_β by exactly T_p: if every member of B_0 was true at w, all still are. Adding a false p lowers it by $|F_p|$, or by $|F_p| + \beta$ if B_0 was error-free at w. So truth-directedness holds. Conservativeness can be imposed on the v_p. The measure does not depend on the current state, so Maximize Expected Accuracy has a single, sequence-independent verdict and Sequence-Irrelevance holds. Every belief state is an option, so Accuracy First holds. I make the following permission bridge explicit:

(Permission) If a belief state uniquely maximizes expected accuracy, it is rationally permitted.

This is an explicit bridge from optimization to permission. A merely necessary accuracy constraint, combined with unspecified further prohibitions, would not license that inference. The counterexample concerns an accuracy-first theory that permits its unique optimum.

The case. Let $W = \{w_1, w_2, w_3\}$, $T_p = 1$ and $F_p = -2$ for every p (so $\tau = 2/3$), and $\beta = 3$. Write $\mathbf{1} = \{w_1\}$. At $C = (0.70, 0.15, 0.15)$ the unique optimum among all 2^8 belief states is $\{\mathbf{1}, \{w_1, w_2\}, \{w_1, w_3\}, W\}$, with expected accuracy 4.30 against a runner-up of 4.20. At $C' = (0.75, 0.01, 0.24)$ the unique optimum is $\{\{w_1, w_3\}, W\}$, with 4.94 against 4.75. Hand computation agrees with exhaustive enumeration in exact arithmetic.[2] So $\mathbf{1}$ is believed at credence 0.70 and not at credence 0.75. By Permission both optima are rational. Proposition 5 would then require $\tau_{\mathbf{1}} \leq 0.70$ and $\tau_{\mathbf{1}} \geq 0.75$, which is impossible.

This is not a knife-edge.

Theorem 1. Let $T > 0 > F$ with $T < -F$, let $\beta > 0$, let $D = T - F$ and $\tau = -F/D$, and put

$$c_0 = \frac{\beta - T - 2F}{2D + \beta}, \qquad c_1 = \frac{\beta - 2F}{2D + \beta}.$$

Then $\max(c_0, \tau) < c_1 < 1$. For all $c < c'$ in the interval $(\max(c_0, \tau), c_1)$, $\mathbf{1}$ belongs to the unique optimum at $(c, \frac{1-c}{2}, \frac{1-c}{2})$ and to no optimum at $(c', \varepsilon, 1 - c' - \varepsilon)$, for all sufficiently small $\varepsilon \geq 0$.

The proof is in the Appendix; it was also checked by enumeration for premiums from 0.01 to 1000. The mechanism is visible in the case. At C the residual 0.30 is spread over two worlds, and the hypothesis $\mathbf{1}$ is the best "core": believing everything it entails earns three positive separable terms and a premium on 0.70. At C' almost all the residual sits on w_3, so the disjunction $\{w_1, w_3\}$ is nearly certain. Its premium on 0.99 outweighs what the more specific hypothesis adds, so the agent hedges. For $0 < \beta \leq D$, the interval has length $\beta T/(D(2D + \beta))$; it narrows as the premium approaches zero, but never closes. This is a

[2] All 256 states over the eight propositions of a three-world space (including $\emptyset$ and W) were enumerated with rational arithmetic. A scan of credences on a grid of mesh $1/40$ found monotonicity violations for all six contingent propositions.

statement that every premium has some counterexample, not that every credence changes its optimal beliefs under an arbitrarily small premium.

The result instantiates Holguín's Non-Monotonicity: a lower credence in a proposition can permit thinking it while a higher credence does not (2022, §5). His argument against Lockeanism used this principle. The record measure shows that accuracy-first epistemology without Separability is not bound by monotonicity either.

4. Diagnosis

Where does the derivation fail? There are two gaps.

The quantifier gap. Truth-directedness gives, for each world w, some positive increment. Dorst's passage on p. 193 then names one number $T_p^{B_0}$ as "the" increment. That moves from "for every w there is a t" to "there is a t for every w." Under the record measure the step fails. Adding the false **1** to $\{\{w_1, w_3\}\}$ costs 2 at w_2 but 5 at w_3, since only at w_3 was the prior state error-free. In expectation, the comparison that Proposition 4 makes between B and $B \setminus \{p\}$ becomes

$$EA_C(B) - EA_C(B \setminus \{p\}) = e_p - \beta\, C(\textstyle\bigcap(B \setminus \{p\}) \setminus p).$$

Belief in p is therefore worth adding iff e_p exceeds a term fixed by the credence of the rest of the core outside p. That is a threshold on $C(p)$ that moves with credences in other propositions, not one fixed by values. What the step needs is:

(ME) Marginal extensionality. For every B_0 and $p \notin B_0$, $A(B_0 \cup \{p\}, w) - A(B_0, w)$ depends on w only through whether $w \in p$.

This is not a consequence of A's being well-defined. It is a substantive claim: the marginal value of a belief may depend on the world only through that belief's own truth value.

The content gap. Grant ME. Proposition 4 then yields thresholds $\tau_p^{B^-}$ indexed to B^-, the rest of the belief state. The proof of Proposition 5 generalizes from "any" B^- to all rational states. That step requires $\tau_p^{B^-}$ not to vary with B^-. Sequence-Irrelevance cannot supply this, because B^- is not a sequence; it is the content of the other beliefs, which is exactly what the holist says matters. For any fixed measure, Maximize Expected Accuracy compares whole states, and its verdicts are already sequence-independent. Dorst himself distinguishes holism from dependence on the order of consideration (2019, p. 195). That is why a principle about order cannot exclude the holism of content. A sufficient strengthening is:

(BI) Background-independence. $T_p^{B_0}$ and $F_p^{B_0}$ do not depend on B_0, for states B_0 not containing the complement of p.

Together, these two assumptions recover separability on the stated domain. BI is stronger than simply requiring the ratio that fixes a threshold to be background-independent. The theorem characterizes this sufficient repair, not every possible repair.

Theorem 2. Let A be truth-directed. Consider only states containing no proposition together with its complement.[3] (a) A satisfies ME iff $A(B, w) = A(\emptyset, w) + \kappa(B) + \sum_{p \in B} J_p\,[w \in p]$ for some world-independent κ and constants $J_p > 0$, the sum run-

[3]The proviso is needed, since there are ME measures in which the marginal value of p changes when its complement is believed. It is harmless here. Under Variable Conservativeness $\tau_p + \tau_{\neg p} > 1$, so such states are never optimal for separable measures (Dorst's Proposition 3), nor, by Theorem 3 below, for the record measure.

ning over the contingent $p \in B$. (b) A satisfies ME and BI iff κ is additive, that is, iff $A(B, w) = A(\emptyset, w) + \sum_{p \in B} v_p(w)$ for some proposition-dependent values v_p.

Since $A(\emptyset, w)$ is the same for every state, it never affects which states are optimal. Part (b) therefore says that adding ME and BI returns us to the separable measures covered by Dorst's Corollary 2. It does not classify every measure whose optima happen to be Lockean. The proof (Appendix) turns on one observation. Adding p and q in either order gives the same total, so $\Delta_p^B - \Delta_p^{B \cup \{q\}} = \Delta_q^B - \Delta_q^{B \cup \{p\}}$, where Δ_r^B is the increment of adding r to B. Under ME the left side is affine in $[w \in p]$ and the right side affine in $[w \in q]$. Unless q is p's complement or trivial, both sides must be constant. A linear-algebra computation over all 256 states on three worlds confirms (a), and it confirms that the complement proviso cannot be dropped.[4]

Part (a) shows what survives of holism under ME: a world-independent term $\kappa(B)$, a bonus or malus for a state's shape regardless of how the world is. A nonadditive κ can make Proposition 5 fail. For example, a small bonus for deductively closed states, small enough to preserve truth-directedness, shifts the effective threshold of p according to whether adding p completes or breaks closure.[5] Dorst can reject such a κ as non-alethic. Rejecting such terms would require a further account of which forms of holistic value qualify as accuracy.

Theorem 2 concerns the derivation, not its conclusion. Some non-separable measures do have Lockean optima. A separable measure lowered on states that are never optimal is an example. So Separability is not necessary for Lockean outputs. What the combined results show is narrower: the proposed inference is not secured by well-definedness and order independence alone. The record measure supplies a truth-directed counterexample, while ME plus BI supplies one transparent repair.

5. What holism delivers instead

The record measure does not leave belief unstructured. Its optima have an exact description. Write $\bigcap B$ for the core of B, and for each proposition X let

$$\Phi(X) = \sum_{p \supseteq X} \max(e_p, 0) + \beta\, C(X).$$

Theorem 3. For any finite W and any proposition-dependent values, $\max_B EA_C(B) = \max_X \Phi(X)$. A state B is optimal iff its core maximizes Φ, B contains every $p \supseteq \bigcap B$ with $e_p > 0$, and B contains no p with $e_p < 0$. In particular, when values are uniform, no e_p is zero, and Φ has a unique maximizing core X^*, the unique optimum is

$$B^* = \{p \supseteq X^* : C(p) > \tau\},$$

where X^* maximizes Φ, and: (i) B^* is consistent iff $X^* \neq \emptyset$, and it is closed under single-premise entailment; (ii) X^* contains every world more probable than any world it contains; (iii) B^* is deductively closed iff $C(X^*) > \tau$, in which case it consists of exactly the consequences of X^*.

[4] The ME constraints form a homogeneous linear system on the 768 values $A(B, w)$. Each vector of a basis of its solution space was tested against the representation in (a), with numerical deviation below 10^{-14}.

[5] With $T = 1$, $F = -2$ and a bonus of $1/2$ for closed states, the measure satisfies ME and truth-directedness. On three worlds, $\{w_1\}$ is believed at $(0.575, 0.2, 0.225)$ but not at $(0.65, 0, 0.35)$.

Call this core-relative Lockeanism. The agent selects a working hypothesis X^*, trading its probability (the premium term) against the number of above-threshold propositions it entails. She then believes those above-threshold propositions that the hypothesis entails. Lockean belief is the special case in which the best core is the intersection of all above-threshold propositions. By (ii) the working hypothesis is always a set of most probable worlds, which is the filtering structure guess-based theories posit for answers (Holguín 2022, §10). By (iii) closure is not guaranteed. On four worlds with $C = (0.62, 0.18, 0.10, 0.10)$ and the values of §3, the unique optimum believes $\{w_1, w_2\}$ and $\{w_1, w_3\}$ but not $\{w_1\}$. The theorem was checked against exhaustive enumeration on three and four worlds, with uniform and with randomly varying values.[6]

Three consequences matter for the debate.

First, accuracy-first epistemology is neutral between Lockean and guess-like belief. Separable value gives thresholds. Record value gives a best hypothesis and its probable consequences. Accuracy does not choose between them; a view about the value of an error-free body of belief does.

Second, the record measure captures both forms of non-monotonicity in the literature. In §3 the agent hedges when a disjunction becomes nearly certain. In Holguín's horse race, belief in A is lost when a rival overtakes it: told that A, B and C have chances 0.40, 0.35 and 0.25, one may think A wins; told 0.45, 0.50 and 0.05, one should think B wins and not A (2022, §5). A single context reproduces both verdicts. Let values depend only on how many horses a proposition mentions: $(T, F) = (1, -0.66)$ for one horse and $(0.65, -1.01)$ for two, which satisfies Dorst's Variable Conservativeness, and set $\beta = 0.19$. The unique optimum at the first credences is the set of consequences of "A wins". At the second it is the set of consequences of "B wins". The runner-up there is the inconsistent state that believes both A and B, which is exactly what a Lockean with these thresholds (about 0.40 for one horse) would believe.[7] So the premium does what Holguín says sophisticated Lockeanism cannot: it keeps thresholds low enough for extreme weakness while ruling out the incompatible beliefs those thresholds would license. The margins are small, though, and a random search suggests that values doing this occupy a narrow band. The result shows that non-monotonicity is available to accuracy-firsters. It does not show that the record measure is the right theory of thinking.

Third, Dorst's reduction of belief to credence (§5.1) survives in a weakened form. Under the record measure, belief can still be "read off" credence and contextual values, since B^* is a function of both. What fails is the Lockean form of the function. The reduction is thus independent of Lockeanism. A Lockean needs some further restriction on the admissible scores or rational states. Separability is one sufficient restriction; it is not shown here to be necessary.

5.1 The cost of an error-free record

The characterization also tells us what increasing the premium buys. Hold the credences, proposition space and individual truth values fixed. A larger premium favors a more probable core, which allows fewer beliefs. This is a different comparison from §3, where the premium was fixed and credences changed.

[6] In 887 random instances with integer credences gave identical predicted and computed optimal sets (799 on three worlds and 88 on four). These finite checks supplement the proof.

[7] Exhaustive enumeration over all 256 belief states, in exact arithmetic, with $T_W = 1$ (the value of believing W does not affect which states are optimal) and $\beta = 19/100$. Expected accuracies: 1.384 against 1.380 at the first credences, and 1.832 against 1.824 at the second.

Theorem 4. Suppose the values are uniform and C has full support and distinct world probabilities. As $\beta > 0$ increases, the probability of any maximizing core cannot decrease. Where the optimum is unique and no $e_p = 0$, the cores expand by inclusion and the belief states shrink by inclusion. For all sufficiently large premiums the unique optimum is $\{W\}$.

For example, if $X \subset Y$, the advantage of the broader core is

$$\Phi_\beta(Y) - \Phi_\beta(X) = \beta C(Y \setminus X) - \sum_{p \supseteq X,\ p \not\supseteq Y} \max(e_p, 0).$$

Broadening the core gains probability for the perfect record and loses individually profitable beliefs. The theorem makes that tradeoff global. Increasing concern for avoiding any falsehood eventually suppresses every contingent belief, even a highly probable one. The premium has not made a true belief intrinsically bad: adding one still improves the score at every world where it is true. What changes is the expected cost of risking the whole record.

This exposes both the attraction and the limitation of the positive model. A researcher who needs jointly reliable premises for a long deduction has a reason to care about their conjunction. But valuing that conjunction without limit makes inquiry vacuous. The score therefore requires a defended balance between acquiring truths and protecting a body of beliefs. Calling both aims accuracy does not fix their relative weight. Nor does the model prove that an ordinary believer should adopt the particular perfect-record bonus used here.

The opposite limit matters too. At a fixed credence with no zero individual scores, the separable optimum is unique. Because there are finitely many belief states, its expected-score advantage has a positive minimum. Every sufficiently small premium preserves it. Holism therefore need not produce dramatic changes throughout the credence simplex. Theorem 1 instead blocks a universal representation claim: however small a positive premium, some credences lie sufficiently close to a tradeoff for the distribution of probability outside a proposition to affect whether it is believed.

The philosophical disagreement can now be stated without an intuition contest about the example. The Lockean localizes the decision about each proposition: once its credence and contextual values are fixed, the allocation of the remaining probability is irrelevant. The record model rejects that independence because the proposition joins a body of beliefs whose joint reliability matters. To derive Lockeanism from accuracy, one must justify that localization, or another restriction that entails the same verdicts. Order independence alone addresses a different issue.

6. Objections

The premium is not accuracy; it double-counts the conjunction. The premium is alethic, since it depends only on which believed propositions are true. It is not a disguised belief in the core: in the four-world case it accrues on $\{w_1\}$, which is not believed. And it respects strict truth-directedness, the only constraint Dorst places on holistic measures. The objector may insist that genuine accuracy is local, so that a belief's contribution depends only on its own truth. That insistence is ME. Coupled with BI it yields separability, but ME alone leaves the additional term identified in Theorem 2(a). To exclude the premium on those grounds is to deny the holism that §4.2 set out to accommodate. The reply leaves one thing open: someone may find the premium unmotivated as a value. Then the dispute is about the value of an error-free record. That is my thesis.

Dorst meant only belief-dependence: values that vary with what else is believed, not with whether it is true. The text resists this reading by claiming that well-definedness alone supports the step (2019, p. 193). Grant the reading anyway. ME then becomes an explicit premise, and the content gap remains. Under ME, Theorem 2(a) leaves a world-independent κ. Nonadditive choices of κ can defeat a fixed threshold; additive ones recover separability. This alternative reading narrows the admissible holism but does not, without further argument, establish the advertised conclusion.

Sequence-Irrelevance should be read strongly, as requiring background-independent thresholds. This is a substantive alternative reading, but it goes beyond invariance under reordering an exam. Background-independent thresholds need not be identified with background-independent marginal values. Theorem 2 concerns the latter, stronger condition. A defender who imposes threshold independence directly owes an argument for why an accuracy holist must accept it; appealing only to order independence leaves the issue unsettled.

Maximize Expected Accuracy gives only a necessary condition; perhaps neither optimum is rational. Then rationality would be unsatisfiable at C and C'. Accuracy First would then have yielded a norm that no state can meet, which no accuracy-firster accepts. Permission is the minimal premise needed for the framework to issue verdicts at all.

The premium is representation-dependent. The value of a record may change when the proposition space is refined: the additive part counts more propositions, while the premium is paid once. This is a serious objection to treating the measure as a complete account of epistemic value. The counterexample fixes a finite proposition space, the same unrestricted space used in the target derivation, and compares two credences on it without changing the questions or their values. Its nonmonotonicity therefore does not arise from switching representations between the two cases. But nothing proved here establishes invariance across different descriptions of the same inquiry. A positive theory based on record value would need to say which questions count and how their weights survive refinement. The limited negative result does not depend on having solved that further problem: the proposed derivation must either exclude the score by a stated premise or accommodate its verdicts. Calling that exclusion a representation-invariance requirement would make the substantive restriction visible.

The result is negative only; Dorst's picture survives. In part, yes. Belief remains a function of credence and contextual values, and Dorst's "best guess" gloss fits core-relative Lockeanism at least as well as Lockeanism. What does not survive is the claim that accuracy forces the Lockean form. That matters to Holguín's argument, which targets the Lockean form. Non-monotonicity tells against sophisticated Lockeanism, but not against accuracy-first epistemology as such.

7. Conclusion

The record measure gives a precise obstruction to the claimed inevitability of Lockean belief. It evaluates every belief state, rewards true additions, penalizes false additions, and judges final states without regard to formation order. With the explicit permission bridge, it nevertheless licenses a nonmonotone relationship between credence and belief for every positive premium. Its optimum has an exact core-and-threshold description; consistency and closure require the additional conditions stated in Theorem 3.

The diagnosis is deliberately limited. ME plus BI restores separability on the specified domain, but separability has not been proved necessary for every Lockean output. Nor has

the record premium been established as the uniquely correct measure of accuracy. What changes is the burden of argument: a defense of Lockeanism must justify a restriction that excludes this kind of holistic value, rather than taking order-independent accuracy maximization to have excluded it already.

Appendix: Proofs and checks

Proof of Theorem 2. Write $\Delta_p^B(w) = A(B \cup \{p\}, w) - A(B, w)$ for $p \notin B$. Under ME, $\Delta_p^B = F_p^B + J_p^B[w \in p]$ for contingent p, and Δ_p^B is constant for $p \in \{\emptyset, W\}$; truth-directedness gives $F_p^B < 0 < F_p^B + J_p^B$. (i) Let $p, q \notin B$ be distinct, with q not the complement of p. Computing $A(B \cup \{p, q\}, \cdot) - A(B, \cdot)$ in both orders gives $\Delta_p^B - \Delta_p^{B\cup\{q\}} = \Delta_q^B - \Delta_q^{B\cup\{p\}}$. The left side is $\alpha + (J_p^B - J_p^{B\cup\{q\}})[w \in p]$ and the right side is $\gamma + \delta[w \in q]$, or a constant if q is trivial. If p, q are contingent, distinct and not complementary, then $1, [w \in p], [w \in q]$ are linearly independent, so $J_p^{B\cup\{q\}} = J_p^B$. If q is trivial, the same conclusion follows. (ii) Building any B that contains neither p nor its complement from $\emptyset$ one member at a time, (i) gives $J_p^B = J_p^\emptyset =: J_p$. (iii) For B without complementary pairs, add its members in any order. Each increment is $F_p^{B_i} + J_p[w \in p]$, so $A(B, w) - A(\emptyset, w) = \kappa(B) + \sum_{p\in B} J_p[w \in p]$, where $\kappa(B)$ is the sum of the F-terms (and the constant increments). It is world-independent, and it is order-independent because the total is. The converse of (a) is immediate. For (b), BI says $F_p^B = \kappa(B \cup \{p\}) - \kappa(B)$ is independent of B, which holds iff $\kappa(B) = \kappa(\emptyset) + \sum_{p\in B} f_p$; then $v_p = f_p + J_p[w \in p]$. Conversely, separable measures satisfy ME and BI trivially. □

Proof of Theorem 3. $EA_C(B) = \sum_{p\in B} e_p + \beta C(\bigcap B)$. Every member of B includes $\bigcap B$, so $EA_C(B) \leq \Phi(\bigcap B)$, with equality iff B contains every $p \supseteq \bigcap B$ with $e_p > 0$ and no p with $e_p < 0$. Conversely, given X, let $B_X = \{p \supseteq X : e_p > 0\}$ and $X' = \bigcap B_X \supseteq X$. The members of B_X are exactly the positive-e propositions including X', so $EA_C(B_X) = \Phi(X') \geq \Phi(X)$, because $C(X') \geq C(X)$. Hence $\max EA = \max \Phi$, and the optimality condition follows. For uniform values, upward closure holds because $p \subseteq q$ implies $C(q) \geq C(p)$. For (ii): if $u \in X^*$, $v \notin X^*$ and $C(v) > C(u)$, swapping u for v maps the supersets of X^* bijectively onto those of the new core without lowering any credence, and it raises $C(X^*)$; that contradicts maximality. For (iii): B^* is upward closed with $\bigcap B^* = X^*$, so it is closed under conjunction iff $X^* \in B^*$, iff $C(X^*) > \tau$. Consistency holds iff $\bigcap B^* \neq \emptyset$. □

Proof of Theorem 1. Write $e_p = C(p)D + F$; note $\tau > 1/2$. At $C_c = (c, \frac{1-c}{2}, \frac{1-c}{2})$ with $c > \tau$, the positive terms are $e_{\mathbf{1}}$, $e_{12} = e_{13}$ and $e_W = T$, and all others are negative (their credences are at most $1 - c < 1/2$). Then $\Phi(\mathbf{1}) - \Phi(W) = e_{\mathbf{1}} + 2e_{12} - \beta(1 - c) = c(2D + \beta) + T + 2F - \beta$, which is positive iff $c > c_0$. Also $\Phi(\mathbf{1}) - \Phi(\{w_1, w_3\}) = e_{\mathbf{1}} + e_{12} - \beta\frac{1-c}{2} > \frac{1}{2}(\Phi(\mathbf{1}) - \Phi(W))$, which is positive. Next, $\Phi(\mathbf{1}) - \Phi(\{w_2\}) = e_{\mathbf{1}} + e_{12} + \beta\frac{3c-1}{2} > 0$ and $\Phi(\mathbf{1}) - \Phi(\{w_2, w_3\}) = e_{\mathbf{1}} + 2e_{12} + \beta(2c - 1) > 0$; the cases of $\{w_1, w_2\}$ and $\{w_3\}$ are symmetric; and $\Phi(\mathbf{1}) - \Phi(\emptyset) = \beta c > 0$. By Theorem 3 the unique optimum has core $\mathbf{1}$ and contains $\mathbf{1}$. At $C' = (c', 0, 1 - c')$ with $\tau < c'$: $e_{\mathbf{1}} = e_{12} = c'D + F > 0$ and $e_{13} = e_W = T$, so $\Phi(\{w_1, w_3\}) - \Phi(\mathbf{1}) = \beta(1 - c') - 2e_{\mathbf{1}}$, which is positive iff $c' < c_1$, while $\Phi(\emptyset) = \Phi(\mathbf{1}) - \beta c'$. A state containing $\mathbf{1}$ has core $\mathbf{1}$ or $\emptyset$, so no optimum contains $\mathbf{1}$. These strict inequalities persist for small ε. Finally, $c_1 - c_0 = T/(2D+\beta)$, $c_1 - \tau = \beta T/(D(2D+\beta))$ and $1 - c_1 = 2T/(2D + \beta)$ are all positive. □

Proof of Theorem 4. Write $H(X) = \sum_{p \supseteq X} \max(e_p, 0)$. If X_i maximizes $H(X) + \beta_i C(X)$ and $\beta_2 > \beta_1$, adding the two optimality inequalities gives $(\beta_2 - \beta_1)(C(X_2) - C(X_1)) \geq 0$. Theorem 3(ii) and distinct world probabilities make each maximizing core a prefix of the worlds ordered by probability. Their probabilities therefore order their inclusions; the formula for B^* reverses those inclusions. Finally, $H(W) = T$, and full support makes $1 - C(X) > 0$ for each proper subset. Thus W uniquely maximizes Φ_β whenever

$$\beta > \max_{X \subsetneq W} \frac{H(X) - T}{1 - C(X)}.$$

Theorem 3 then gives $B^* = \{W\}$. At $\beta = 0$, if every $e_p \neq 0$, the finite separable problem has a unique optimum with some gap $g > 0$ over every rival. Changing β from zero changes each pairwise score difference by at most β, so $0 < \beta < g$ preserves that optimum. □

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When Acting on Knowledge Makes Free Evidence Harmful: Good's Theorem in Normality Models

Abstract. Knowledge-based decision theory tells an agent to maximize expected utility with probabilities conditioned on what she knows. I examine this rule inside Goodman and Salow's normality theory of inductive knowledge. In finite statist, comparable models, with K-normality and the specified questions available, it gives every free question non-negative value at a situation if and only if the agent's knowledge is gapped: each possibility she leaves open is sufficiently more normal than each evidential possibility she rules out. More generally, any factive state that leaves open whatever rational belief leaves open can be safely conditioned on only if it is gapped. In graded models a factive gapped state must coincide with the evidence. A separate positive-measure argument extends the negative result to their Gaussian measurement model; at normal situations a second weighing has negative value at ten-to-one stakes. An expected-utility form of Salow's Conditional Knowledge Dominance does not help. In the finite setting it fails wherever the unconditional rule fails, even when it also governs later choices. The counterexamples survive small payoff changes. These results constrain one way of giving knowledge practical authority; they do not refute every knowledge-based theory.

1. Introduction

If you know p, any evidence against p is misleading, so it seems you may avoid it. This is the dogmatism puzzle. Salow (2025) offers the most economical recent dissolution. If Preservation holds (discovering something consistent with your beliefs never requires giving one up), you cannot believe both p and that you might meet evidence defeating p, so the puzzle never arises. If Preservation fails, the puzzle needs Weak Knowledge Dominance, which Salow replaces with a principle that looks equivalent but is not:

> Conditional Knowledge Dominance: If you know that B's consequences would be better than A's if they're not equally good, you should choose B over A. (Salow 2025, 33)

In Salow's Suitcase, you do not know that the suitcase weighs under 18 kg on the hypothesis that it weighs over 17.5 kg, since discovering the latter would defeat that knowledge. So the principle does not tell you to avoid checking. Salow's principles are qualitative. He grants that how all-out attitudes guide choices made under uncertainty about outcomes is "a general problem", and he expects that "Any solution to that problem will generalize to this one" (2025, 34).

The most developed solution of that kind is knowledge-based decision theory (KDT): maximize expected utility using probabilities conditioned on what you know, so that known propositions get probability 1. Goldschmidt (2024) axiomatizes a version due to Hawthorne and Stanley (2008), and Hong (2024) explores the view. I ask whether KDT keeps the practical content of Salow's dissolution, namely Good's (1967) theorem that free evidence never has negative expected value, once knowledge is modeled by the most developed formal theory of inductive knowledge, the normality framework of Goodman and Salow (2023, 2025). My thesis is this:

> In finite statist models with the specified questions available, conditioning on a factive belief-containing state makes every question safe only if that state is gapped. Under K-normality and comparability the condition is also sufficient. The pro-

posed conditional expected-utility repair has counterexamples wherever the unconditional rule does.

Three results support the thesis. First, an exact boundary (Theorem 1). With K-normality, statism and comparability, KDT gives a free question non-negative value for every decision problem if and only if no situation compatible with what one knows is one where answering would reopen a possibility one now rules out. Every question has non-negative value if and only if knowledge is gapped. Theorem 2 extends the necessity half to any factive state that leaves open what rational belief leaves open, whichever theory of knowledge one adopts. Second, ubiquity. In Goodman and Salow's own model of Bjorn weighing himself twice, some bet makes KDT price the free second weighing negatively at each situation. At normal situations the displayed bet already does so at ten-to-one stakes (§3). Third, the conditional repair fails (§5). I define the expected-utility analogue of Conditional Knowledge Dominance. It handles the bets that motivate it. But in any situation where KDT disvalues a question, some decision problem makes the conditional rule disvalue it too (Theorem 3). The proof removes payoff ties both between the primitive acts and between acting now and the later policy. The rule therefore fails even when used consistently at both stages.

The closest prior work is as follows. Salow (2025) shows that Weak Knowledge Dominance licenses dogmatism exactly where Preservation fails, and he calls such failures "relatively rare" (36). I give the expected-utility counterpart, with an exact boundary, and show that in graded models failures of Good arise at every situation with inductive knowledge. I also show that his remedy does not carry over to expected utility. Goodman and Salow (2025, henceforth BRN) show that statism, collapse and comparability validate Preservation and that Preservation fails in Bjorn. I give the decision-theoretic counterpart, together with what follows for decisions from their distinction between conditional and post-discovery belief. Geanakoplos (1989; 2002 revision, §2) gives sufficient conditions for the value of information, discussed by Das (2023). The contribution here is the bridge from normality kens to information value, with an exact safety boundary. The necessity proof directly constructs a harmful question; it does not infer harm merely from failure of sufficient conditions. Hong (2026, §§1–3) directly criticizes normality-based action guidance through lottery switching, sweeteners and the value of otherwise irrelevant information. Those problems already challenge conditioning on normality knowledge. The proposed increment here is an exact finite safety boundary and a construction defeating the specified conditional expected-utility repair, rather than the general claim that normality creates decision-theoretic difficulties.

The result concerns a specific practical role for knowledge: settling probabilities used in expected-utility calculations. It does not deny that knowledge can guide action in other ways, or refute Salow's qualitative principle by itself. Its force depends on whether the theorist wants the universal safety guarantee, a restriction made explicit below.

2. Framework

I follow Goodman and Salow (2023) and BRN. A normality structure has a set S of states, a family $\mathcal{E}$ of possible bodies of evidence, and two relations on situations $\langle s, E\rangle$ (with $s \in E \in \mathcal{E}$): $\succeq$ (at least as normal as), a preorder, and $\gg$ (sufficiently more normal than). $\gg$ implies $\succeq$, is irreflexive, and chains: if $w_1 \succeq w_2 \gg w_3 \succeq w_4$ then $w_1 \gg w_4$. Under statism both relations are evidence-independent, so I treat them as relations on states. Comparability says any two states with the same evidence are $\succeq$-related. Collapse says every strict difference in normality is sufficient. For the theorems S is finite.

Rational belief, given evidence E, rules out exactly the evidential possibilities that something else in E is sufficiently more normal than:

$$B(E) = \{s \in E : \text{ no } u \in E \text{ has } u \gg s\}.$$

Knowledge goes beyond evidence by the same mechanism, but its extent depends on where one actually is. I work mainly with K-normality (Goodman and Salow 2023, 104):

$$K_E(x) = B(E) \cup \{v \in E : v \succeq x\}.$$

One knows what is true throughout $K_E(x)$. Evidence is transparent: situations with the same evidence are evidentially accessible from each other. As Goodman and Salow note, "given transparency, K-normality entails KK" (2023, 107). The alternatives they consider, I- and KM-normality, validate their principle Margins and allow failures of KK; I return to them in §6.1. Discovering p at $\langle x, E\rangle$ yields $\langle x, E \cap p\rangle$. A question is a partition Π of E into discoverable cells; $\Pi(s)$ is the cell containing s.

Knowledge-based expected utility. Fix a prior Pr with full support. A decision problem is a finite set A of acts, each a function from states to utilities. At $\langle x, E\rangle$, KDT chooses an act maximizing $\Pr(u_a \mid K_E(x))$, with ties broken by a fixed order. Goldschmidt (2024) describes this conditionalizing form of KDT alongside Hawthorne and Stanley's proposal, on which known propositions receive probability 1; for present purposes the two agree.

The value of a free question. Let $\alpha_\Pi(s)$ be the act KDT chooses at $\langle s, E \cap \Pi(s)\rangle$, after answering Π in state s. The KDT value of answering Π before choosing is

$$V_x(\Pi, A) = \sum_s \Pr(s \mid K_E(x))\, u(\alpha_\Pi(s), s) \;-\; \max_{a \in A} \Pr(u_a \mid K_E(x)).$$

This is Good's comparison of acting now with the plan of looking and then acting, with $\Pr(\cdot \mid K_E(x))$ in place of the unconditioned probability. Call Π safe at x if $V_x(\Pi, A) \geq 0$ for every decision problem A.

Two conditions will matter.

Relative defeat at x for Π: some $s \in K_E(x)$ has $K_{E\cap\Pi(s)}(s) \not\subseteq K_E(x)$. In words, for all one knows, answering the question would leave open something one now rules out.

Gapped: every member of $K_E(x)$ is sufficiently more normal than every member of $E \setminus K_E(x)$.

3. Bjorn's second weighing

Goodman and Salow's (2023, §6) model of Bjorn is statist and comparable, and it violates collapse. Its states are triples $\langle w, y_1, y_2\rangle$: Bjorn's weight and the readings of two scales. With $\delta = (w - y_1)^2 + (w - y_2)^2$, one state is at least as normal as another iff its δ is no larger, and sufficiently more normal iff its δ is smaller by more than c^2. Goodman and Salow derive these relations from independent Gaussian errors (2023, 129). I take the standard deviation as the unit and, following BRN (§5, n. 18), set $c \approx 1.96$.

After the first reading r, Bjorn believes his weight lies in $[r - c, r + c]$. Under K-normality, in any situation with $\delta \leq c^2$ this belief is knowledge, and his ken is $\{\delta \leq c^2\}$. After a second reading r' he believes his weight lies within $c/\sqrt{2}$ of the average of the two readings. So he leaves open weights up to $c/\sqrt{2} + |r - r'|/2$ from r. As Goodman and Salow show for the case

where the first reading is accurate, he "loses knowledge" whenever $|r - r'| > (2 - \sqrt{2})c$ (2023, 127). Since the readings may differ by up to $\sqrt{2}c$ for all he believes, he believes that the second weighing might defeat his knowledge. This is a failure of Preservation, the case BRN use against it. It is not rare: with Gaussian errors, the second reading falls in the Preservation-violating range with probability about 0.37, as BRN also estimate (§5, n. 18). Salow describes such failures as "relatively rare" (2025, 36); in the leading model of measurement they occur on more than a third of second weighings.

Ten-to-one bet. After the first weighing Bjorn is offered a bet: win 1 if his weight is within c of r, lose 10 otherwise. He knows the bet wins, so KDT takes it. A free second weighing is then offered. After the second weighing he would decline the bet whenever the readings differ by more than about $1.12c$, because his new ken gives the losing weights enough probability. Conditional on his present knowledge, this has probability about 0.05. In each such state he forgoes a payoff he now knows he would get. So KDT values the free weighing at about -0.05. At twenty-to-one the value is -0.12, and at a hundred-to-one -0.27. (These are numerical integrations over the model; see the appendix.) A fallibilist who uses the unconditioned probabilities also takes the ten-to-one bet: on her evidence it wins with probability 0.95, so its expected value is $0.95 - 10(0.05) = 0.45$. By Good's theorem the second weighing cannot have negative value for her.

The phenomenon is not confined to normal situations. At any situation after the first weighing there is a state compatible with Bjorn's knowledge in which the second reading would reopen a weight he now rules out (appendix, A.5). The finite-state Lemma 1 cannot by itself establish that claim in a continuous space, where a single state has probability zero. Appendix A.5 supplies the positive-measure extension. Thus some finite-stakes bet makes the second weighing harmful at each situation; the required stakes can vary with the situation. The ten-to-one example concerns the normal situations specified above. This payoff ratio does not by itself specify the absolute practical stakes.

4. Where Good's theorem holds

The case instantiates a general pattern. Write M for any assignment of a set $M_F(s) \subseteq F$ to situations, used in place of K in the definitions of §2.

Lemma 1 (defeat is costly). If some $s_0 \in M_E(x)$ has $M_{E \cap \Pi(s_0)}(s_0) \not\subseteq M_E(x)$, then some two-act decision problem gives the question Π negative value.

Proof. Take a bet β paying 1 on $M_E(x)$ and $-L$ elsewhere, and declining, which pays 0. Now β has conditional expectation 1, so it is chosen. At s_0 the post-answer state contains a state outside $M_E(x)$, which has positive probability, so for large L the agent declines. She then gets 0 where β paid 1. Nowhere in $M_E(x)$ does she get more than 1. So $V \leq -\Pr(s_0 \mid M_E(x)) < 0$. □

Lemma 2 (laminar sufficiency). Suppose the post-answer states $\{M'(s) : s \in M_E(x)\}$ all lie within $M_E(x)$. Suppose also that they are reflexive ($s \in M'(s)$), transitive ($t \in M'(s)$ implies $M'(t) \subseteq M'(s)$) and nested (any two are disjoint or one contains the other). Then $V \geq 0$ for every decision problem.

This is an application of Geanakoplos's value-of-information result (1989; 2002 revision, Theorem 1), also discussed by Das (2023). I give a short proof in the appendix.

Lemma 3. Under statism and comparability, for each F the K-normality kens $\{K_F(s) : s \in F\}$ are reflexive, transitive and nested.

Theorem 1. Assume statism, comparability and K-normality. (a) A question is safe at x if and only if there is no relative defeat at x for it. (b) Suppose that for each $v \in E$ the question $\{p_v, E \setminus p_v\}$ is available, where $p_v = \{u \in E : \text{not } u \gg v\}$. Then every available question is safe at x if and only if $K_E(x)$ is gapped.

Part (a) combines Lemmas 1–3. Part (b) adds two observations. If the ken is gapped, no answer can reopen anything. If it is not gapped, the question about p_v reopens v. That is the next lemma, and it does not depend on which theory of knowledge one adopts.

Lemma 4 (non-gaps are defeasible). Assume statism. Let $m \in M_E(x)$ and $v \in E \setminus M_E(x)$, with not $m \gg v$. Then $m, v \in p_v$ and $v \in B(E \cap p_v)$.

Proof. $m \in p_v$ by hypothesis and $v \in p_v$ by irreflexivity. Nothing in p_v is sufficiently more normal than v, so $v \in B(E \cap p_v)$. □

Say that M leaves open what rational belief leaves open if always $B(F) \subseteq M_F(s)$. Knowledge on each of Goodman and Salow's three completions does, and so do justified true belief and the closure of knowledge under iteration.

Theorem 2 (only gapped states can be conditioned on). Assume statism. Suppose M leaves open what rational belief leaves open and the questions of Theorem 1(b) are available. If expected utility conditioned on M makes every available question safe at x, then $M_E(x)$ is gapped. A gapped set containing x contains $C(x)$: the states reachable from x by steps, each to a state that is not sufficiently less normal than the last.

The boundary in Theorem 1 is not the boundary of Preservation. At abnormal situations, knowledge can be defeated by evidence that contradicts one's beliefs, so KDT can disvalue a question even in models where Preservation holds without exception. Take three states a, b, c, in decreasing order of normality, with $a \gg b$ and $a \gg c$ but not $b \gg c$. In this structure Preservation holds for every body of evidence and every discovery. At b one knows $\{a, b\}$. Learning that a is false, which is true at b, reopens c. Salow anticipates this pattern: a 'for all you know' version of Weak Knowledge Dominance "can support dogmatism even given Preservation" (2025, 33). Theorem 1 locates its expected-utility counterpart exactly.

Call E graded if $C(x) = E$ for every $x \in E$. Bjorn's evidence after one reading is graded, since a chain of small increases in δ connects any two states. So is the evidence in Flipping for Heads, where flips are counted until the first heads (BRN §5).

Corollary. (i) Under statism, collapse and comparability, every K-normality ken is gapped, so KDT makes every question safe. (ii) If E is graded, the conditioning state is factive, and the questions of Theorem 1(b) are available, expected utility conditioned on a state that leaves open what rational belief leaves open makes every question safe at x only if that state is all of E.

Part (i) is the decision-theoretic counterpart of BRN's result that statism, collapse and comparability validate Preservation. Part (ii) says that in graded models the only belief-containing state one can condition on without making some free question harmful is one's evidence. Conditioning on evidence is what Salow calls the fallibilist response, on which "decisions should be governed by underlying probabilities rather than our knowledge" (2025, 23). Bjorn shows that the questions needed for (ii) can be as natural as a second reading.

I also checked the theorems by exhaustive enumeration. The enumeration covered all statist comparable structures with up to four states, every body of evidence, every situation and every partition: 64,004 situation–question pairs. Across all available partitions, a situation admitted some relative defeat exactly when its ken was not gapped. This is an existential claim over questions: a particular question can be safe at a nongapped ken. In each of the 3,306 cases of relative defeat, the decision problem built in Theorem 3 below gave

negative KDT value. About 240,000 random decision problems without relative defeat gave no negative value.

5. The conditional repair

5.1 The rule

The natural expected-utility analogue of Conditional Knowledge Dominance compares two options by their expected difference on the hypothesis that they differ. Let $D_{a,b} = \{s \in E : u_a(s) \neq u_b(s)\}$. Let $K_E^D(x)$ be what one knows on the hypothesis D. Its K-normality form restricts evidential accessibility to D, as BRN's definition of conditional belief does (App. D.3):

$$K_E^D(x) = \{v \in E \cap D : \text{no } u \in E \cap D \text{ has } u \gg v\} \cup \{v \in E \cap D : v \succeq x\}.$$

Under statism, when $x \in D$, this is the ken one would have on discovering D; in every case it contains the belief one would have on discovering D. So it respects Salow's constraint that one is "entitled to believe q on the hypothesis that p only if discovering p would not require you to stop believing q" (2025, 33).

CEU. Prefer b to a if $\Pr(u_b - u_a \mid K_E^{D_{a,b}}(x)) > 0$. Looking before acting is one option; its payoff in state s is $u(\alpha_\Pi(s), s)$.

For Bjorn's bet CEU works as Salow's diagnosis predicts. Looking and betting now differ exactly where the second reading would make Bjorn decline. On the hypothesis that this happens, his conditional ken includes losing weights. With the actual readings perfectly accurate, at ten-to-one the bet's expected value on that hypothesis is -0.63. So CEU prefers to look. The dogmatic argument fails for the reason Salow gives: on the hypothesis that the evidence makes a difference, you do not know that your verdict is right.

5.2 Why the repair fails

For the displayed bet, the conditional repair uses states where looking and acting now pay the same. These are excluded from D, so conditioning on a difference carries information. That feature can be removed by a carefully chosen perturbation: if the present act and later policy differ in every state, $D = E$, conditional knowledge is ordinary knowledge, and CEU coincides with KDT. The following construction establishes such a case whenever there is relative defeat. No claim about the prevalence of ties in typical decision problems is needed.

Theorem 3. Assume statism, comparability and K-normality, and suppose conditional knowledge on the trivial hypothesis E is ordinary knowledge. If there is relative defeat at x for Π, then some three-act decision problem makes CEU strictly prefer acting now to answering Π, even if CEU also governs every later choice. KDT gives Π negative value too. These verdicts persist on an open neighborhood of the constructed payoff matrix.

Construction. Let $M = K_E(x)$ and choose $s_0 \in M$ witnessing relative defeat. Take the bet β of Lemma 1, declining, and $\beta_1 = \beta + \varepsilon\zeta$, where $\zeta(s_0) = N$ and $\zeta = -1$ elsewhere. Choose N with positive current mean for ζ, $0 < \varepsilon < 1$, and then L large enough that every post-answer ken meeting $E \setminus M$ strictly prefers declining. There are finitely many such kens, so one finite L suffices. Any remaining post-answer ken excludes s_0: otherwise transitivity would put s_0's outside possibilities in that ken too. There β strictly beats β_1 and declining.

Every pair of primitive acts differs at every state: β and β_1 are everywhere nonzero and their difference is everywhere nonzero. Thus, at every decision stage, the hypothesis on

which CEU conditions is the entire current evidence. CEU and KDT consequently choose the same primitive act. This answers a potential objection to the construction: it does not make the conditional rule anticipate successors who follow a different rule.

Looking therefore never selects β_1, while β_1 is uniquely best now. Its payoff also differs everywhere from the looking policy, so their comparison again uses ordinary knowledge. The gain from acting now at s_0 outweighs the small losses elsewhere (Appendix A.6).

For a numerical illustration, take three states $a \succ b \succ c$, with only $a \gg c$, and a uniform prior. At a the ken is $\{a, b\}$. Ask whether a; on the other answer the ken is $\{b, c\}$. Let the acts pay $(1, 1, -10)$, $(0, 0, 0)$, and $(0.9, 1.5, -10.1)$. Both rules choose the third now, with current expectation 1.2. After looking they choose the first at a and decline at b and c. That policy pays $(1, 0, 0)$ and has current expectation 0.5. CEU therefore prefers acting now by 0.7. All comparisons are strict and no relevant pair has an equal payoff at any state.

The example is stable under sufficiently small changes in every payoff. In the general construction, finiteness and strict preferences supply positive margins; small perturbations preserve the chosen acts and the absence of equal payoffs. This is local robustness, not a claim that these problems are prevalent.

A fixed absolute tolerance for a "significant" difference does not rescue the rule. Scale all utilities by a sufficiently large positive constant so that every relevant pointwise difference exceeds that tolerance. Expectations and choices scale together. This argument addresses a fixed absolute tolerance; it does not exclude context-sensitive or relative notions of significance. Nor does the finite construction establish the corresponding CEU theorem for continuous Gaussian models, where arbitrarily small outside probabilities can prevent a uniform choice of L.

The result also applies to any pairwise rule that agrees with knowledge-conditioned expected utility whenever the compared payoffs differ everywhere. The difficulty is therefore precise: a repair that changes only the treatment of equal-payoff states cannot guarantee safety in these models. It must also reconsider some comparisons on which there are no equal-payoff states to remove.

5.3 Which conditional knowledge?

Outside statist models, conditional knowledge presents a further choice. BRN distinguish belief on a hypothesis, evaluated using the current normality relations, from belief after discovering that hypothesis, when those relations may change (App. D.3). The distinction matters because Salow's reply requires a connection to what discovery would defeat. I do not assume that BRN's treatment uniquely determines a theory of conditional knowledge. Theorem 3 avoids this interpretive question: it uses only the requirement that conditioning on the entire evidence leaves ordinary knowledge unchanged. A non-statist or differently defined conditional theory requires separate assessment.

6. Objections and replies

6.1 "This is self-trust, not evidence"

The objection says that KDT disvalues questions because the agent distrusts her future self, not because she distrusts the evidence. On the Margins completions (I- and KM-normality), statism, comparability and Margins "together entail no defeat" (Goodman and Salow 2023, 128). There, by Lemma 4, the reopened possibility lies in the agent's own present ken at m, so the relative defeat in Theorem 2 is a failure of KK at x. Such failures harm the empty

question too. By Lemma 1 with the trivial partition, a KDT agent without KK at x pays to decide now rather than later on the same evidence. This kin of Williamson's unmarked clock is discussed by Ahmed and Salow (2019) and Das (2023). For the Margins views I concede the objection: what fails there is deference to one's later self, not openness to evidence.

For K-normality the objection fails. KK holds, so by Lemmas 2 and 3 the empty question always has nonnegative value. Zero is not guaranteed: at an abnormal situation the present ken may contain states with strictly smaller kens, whose state-dependent choices can outperform one fixed act. For example, take a more normal than b, with $a \gg b$, a uniform prior, and acts paying $(1, 0)$ and $(0, 2)$ at (a, b). At b the present best value is 1; with no new evidence, the state-dependent choice plan pays $(1, 2)$ and has value 1.5. The empty question therefore has value 0.5. The relevant contrast is absence of a negative value, not universal equality to zero. Moreover, if a question has $V_x(\Pi, A) < 0$ while the empty question has nonnegative value, it also fares worse than the matched plan of waiting without gathering evidence. The negative comparison is not produced merely by contrasting later state-dependent choice with a single present act. In Bjorn's case, the new reading destroys knowledge despite positive introspection. The structure is the classic dogmatist's. I know the bet wins. In some states compatible with what I know, the evidence will make me unable to rely on that. So I should avoid the evidence. Under Theorem 1's assumptions, some available question has this structure exactly where knowledge is not gapped. So the dilemma for graded inductive knowledge is this. On the KK-friendly completion, KDT avoids evidence. On the KK-unfriendly completions, it avoids deferring to itself.

6.2 "Trivial once stated: knowledge is treated as certainty"

The pieces are indeed familiar. Lemma 1 is an ordinary bet, Lemma 2 is Geanakoplos's sufficiency condition, and Corollary (i) parallels BRN's result. The proposed contribution has three parts. The first is the exact location of the boundary, which includes abnormal situations whose kens extend below the belief set. The second is Theorem 2's completion-independence: no belief-containing state other than a gapped one can be conditioned on, so no refinement of knowledge (iterated knowledge, justified true belief) escapes. The third is Theorem 3. Harman's and Kripke's discussions explain why knowledge-as-certainty invites the puzzle. They do not show that the leading conditional remedy fails in expected-utility form. The necessity result concerns fixed conditioning states and the stated universal safety requirement, not every response to dogmatism.

6.3 "KDT is meant for low stakes"

Under K-normality with transparency, iterating knowledge does not change the ken, so conditioning on higher-order knowledge gives the same probabilities here. A ten-to-one payoff ratio is not an absolute measure of stakes. At a fixed ken, positive rescaling of all utilities preserves choices and the sign of information value, so the counterexample can have arbitrarily small payoff magnitudes. But if the ken itself changes with the menu or practical stakes, the theorem must be applied to that changed model. It does not refute every stakes-sensitive theory.

6.4 "Statism is the exception"

BRN argue that statism fails in general. The finite equivalence is conditional on statism, and it should not be advertised as a classification of their entire framework. Bjorn's measure-

ment model is statist, however, so that restriction leaves a substantive application. Beyond it, the relevant questions are whether the post-answer kens remain nested and whether the questions used in Lemma 4 still reopen the excluded possibilities. Neither condition should be assumed without proof.

6.5 Encroachment

Salow (2025, 24 n. 3) reports Hawthorne's (2004, 181) suggestion that, given Good's theorem, the puzzle is handled by pragmatic encroachment of the kind on which encroachment occurs whenever the difference between maximal and actual probability affects one's preferences over the available options. Salow objects that encroachment would not be confined to failures of Preservation (36). Theorem 2 makes the cost exact. To guarantee safety for every decision problem and every question in the theorem's class on graded evidence, a factive belief-containing state must expand to include the entire evidence set. This does not show that the mere availability of a second scale destroys all inductive knowledge at every fixed stake level. The universal guarantee quantifies over decision problems as well as questions.

6.6 "Good's theorem is not sacrosanct"

Ahmed and Salow (2019) contest unrestricted requirements to gather and use information. Bjorn's readings partition the evidential possibilities exactly, but the knowledge kens used for decisions can still overlap. Thus exact observation does not by itself put the decision problem back inside the classical partitional theorem. The negative result is a cost for a theorist who wants universal safety, not a contradiction for one who rejects it.

Another response allows the agent to commit to ignoring an answer. If that commitment is feasible, the menu of policies includes the present best act, so the best available policy cannot be worse merely because a free signal is offered. The results concern looking followed by the specified future choice rule, without such commitment. Their practical significance is the resulting tension between present evaluation and later rational choice. Whether knowledge should determine numerical probabilities, whether the later rule should change, or whether such tensions are acceptable remains a philosophical choice.

7. Conclusion

In finite statist, comparable models with K-normality and the specified questions available, expected utility conditioned on knowledge makes every free question safe exactly where knowledge is gapped. Any factive state that leaves open what rational belief leaves open needs the same condition. On graded normality, which Goodman and Salow defend for measurement, that condition means not going beyond the evidence. The expected-utility form of Salow's Conditional Knowledge Dominance does not escape: in the finite setting it has counterexamples wherever the unconditional rule does, including when the conditional rule governs both stages. What has not been shown is that no other way of letting all-out attitudes guide graded choice could preserve Salow's dissolution. The result places a precise burden on this conditionalizing form of knowledge-first decision theory. It must weaken the universal safety requirement, restrict its model or question class, or change how knowledge controls numerical choice. Collapse is one sufficient restriction, and evidence-based probabilities are one alternative; the results do not establish an exhaustive trilemma for every knowledge-first theory.

Appendix: proofs and computations

A.1 Lemma 2. Let $\mathcal{F} = \{M'(s) : s \in M\}$, $M = M_E(x)$. By nestedness and reflexivity the inclusion-maximal members of $\mathcal{F}$ partition M. For a maximal $C = M'(c)$, let a_C be the act chosen on C. I show by induction on $|C|$ that $\Pr(u(\alpha(s), s) \mid C) \geq \Pr(u_{a_C} \mid C)$. By transitivity every $s \in C$ has $M'(s) \subseteq C$. The states with $M'(s) = C$ choose a_C. The others lie in the maximal proper sub-kens $D_1, \ldots, D_k$, which are disjoint members of $\mathcal{F}$ satisfying the hypotheses. By induction, on each D_j the plan does at least as well as the best act on D_j, and so at least as well as a_C. Summing over the parts of C and then over the maximal sets, the plan's expectation on M is at least $\sum_C \Pr(C \mid M) \max_a \Pr(u_a \mid C) \geq \max_a \Pr(u_a \mid M)$. □

A.2 Lemma 3. Reflexivity holds because $s \succeq s$. For transitivity, let $t \in K_F(s)$. If $t \in B(F)$, then every $v \succeq t$ is in $B(F)$: if $u \gg v$, chaining gives $u \gg t$. So $K_F(t) = B(F) \subseteq K_F(s)$. If $t \succeq s$, then $\{v : v \succeq t\} \subseteq \{v : v \succeq s\}$. For nestedness, by comparability the sets $\{v : v \succeq s\}$ are linearly ordered by inclusion, and each ken is their union with $B(F)$. □

A.3 Theorem 1. (a) Answers in different cells give disjoint kens. By Lemma 3 the post-answer kens are reflexive, transitive and nested. Without relative defeat they lie within $K_E(x)$, so Lemma 2 applies. Conversely, use Lemma 1. (b) Suppose the ken is gapped, $s \in K_E(x) \cap c$, and $v \in c \setminus K_E(x)$. Then $s \gg v$, so $v \notin B(E \cap c)$. Also not $v \succeq s$, since otherwise $v \succeq s \gg v \succeq v$ gives $v \gg v$. So $K_{E \cap c}(s) \subseteq K_E(x)$. If the ken is not gapped, Lemma 4 gives relative defeat, because $B(E \cap p_v) \subseteq K_{E \cap p_v}(m)$. □

A.4 Theorem 2. Lemmas 4 and 1 give the first claim. If M is gapped, $m \in M$ and not $m \gg v$, then $v \in M$; so M is closed under the steps that generate $C(x)$. □ Corollary (i). Let $m \in K_E(x)$ and $v \notin K_E(x)$. Then not $v \succeq x$, and some $u \gg v$. If $m \succeq x$, comparability and collapse give $x \gg v$, hence $m \gg v$. If $m \in B(E)$, then $v \succeq m$ would give $u \gg m$, so m is strictly more normal than v and collapse gives $m \gg v$. □

A.5 Bjorn and positive measure. After the first reading, use coordinates $\mu = (e_1 + e_2)/2$ and $d = e_1 - e_2$, so $\delta = 2\mu^2 + d^2/2$. For the numerical model assume the proper conditional Gaussian measure with independent $\mu \sim N(0, 1/2)$ and $d \sim N(0, 2)$, corresponding to the swamped-prior treatment of weight. The measurement likelihood alone would not fix this conditional distribution without that prior assumption. Conditional on a second reading, d is fixed and the post-answer ken has $|\mu| \leq \max(c/\sqrt{2}, |\mu_s|)$.

Let $R = \max(c^2, \delta(x))$, so the current ken is the disk $M = \{\delta \leq R\}$. Choose an interior point with $\mu_s = 0$ and $d(s)^2/2$ strictly between $R - c^2$ and R. Its post-answer ken reaches beyond M, since its maximum δ is $c^2 + d(s)^2/2 > R$. The inequalities are strict. Hence a sufficiently small open neighborhood U of that point stays inside M and has post-answer conditional probability of M^c bounded below by some $\eta > 0$. Gaussian density makes U have positive current probability. A bet paying 1 on M and $-L$ outside it is therefore declined throughout U when $L > (1 - \eta)/\eta$. It is accepted now; subsequent choice never pays more than 1 inside M. Its value of looking is at most $-\Pr(U \mid M) < 0$. This is the continuous argument that the finite singleton proof cannot supply.

For the stated normal-situation figures, the bet is instead on $|e_1| \leq c$. Integrating the Gaussian density over $\delta \leq c^2$ gives the probabilities of declining after the second reading. These figures are numerical approximations, not premises of the finite theorems.

A.6 Theorem 3. Put $P = \Pr(\cdot \mid M)$ and choose N with $NP(s_0) > 2$. Fix $0 < \varepsilon < 1$. For each post-answer ken F meeting $E \setminus M$, write $q_F = \Pr(E \setminus M \mid F) > 0$. The expected payoff of β_1 on F is at most $1 + \varepsilon N - Lq_F$, and that of β is at most $1 - Lq_F$. Since there are finitely

many F, choose L making both bounds strictly negative. On every other post-answer ken, transitivity excludes s_0, so $\beta_1 = 1 - \varepsilon$ and $\beta = 1$. At the present ken β_1 is uniquely optimal since $P(\zeta) > 0$. At s_0 acting now gains $1 + \varepsilon N$; elsewhere inside M looking gains at most ε. Hence

$$P(u_{\beta_1} - u_{\text{look}}) \geq P(s_0)(1 + \varepsilon N) - \varepsilon > 0.$$

Outside M, factivity forces declining after looking, whereas $\beta_1 = -L - \varepsilon$. Inside M, the looking payoff is 0 or 1, neither equal to β_1. Thus the comparison's difference event is E. The primitive acts likewise differ everywhere, proving CEU–KDT agreement at both stages. Finally, finitely many strict choice inequalities, nonzero pointwise differences, and the strict preference for acting now persist on a sufficiently small open neighborhood of the payoff matrix. □

Computations. Python scripts enumerate all statist structures satisfying Goodman and Salow's constraints: comparable ones with up to four states (up to five for collapse) and non-comparable ones with up to three. For each they compute kens, post-answer kens, KDT and CEU choices with random full-support priors and the constructed problems. Bjorn's values are computed by numerical integration. Flipping for Heads is computed exactly, truncated at 60 flips.

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Limits of Uniform Normality Filters in Actual Causation

Abstract. Normality can filter the counterfactual worlds that witness actual causation. I classify uniform componentwise rankings based on default status and equation conformity: four calibrating judgments hold exactly when the ranking satisfies a license rule, but no ranking separates three further cases under any of four specified default codings. The limitation extends to compensating scores. Under the modified Halpern–Pearl definition, every uniform additive score preserving the chain and ducking judgments counts the switch in a binary Doorbells model as a cause. An exact identity among witness-score changes proves this; an explicit counterexample shows why the restriction to the modified definition matters. These results identify costs of transferring one normality policy across cases. They extend an established discussion of normality and representation, rather than refuting richer normality theories.

1. Introduction

Two stories with the same structural equations and actual values can draw opposite causal verdicts. A bodyguard's antidote in coffee that nobody poisoned is not a cause of the victim's survival, yet the model of that story has the same form as a model of two overdetermining causes. Halpern and Hitchcock (2015) call this the problem of isomorphism, and one response they develop adds a normality ordering on worlds: a counterfactual world witnesses causation only if it is at least as normal as the actual world. Andreas and Günther (2024) add a deviancy condition to a regularity theory for the same purpose, and Gallow (2021) builds the default/deviant distinction into what a cause is.

Another familiar contrast is between switching and early preemption. A flipped switch sends a train down the right-hand track instead of the left, the tracks reconverge, and the flip is not usually counted a cause of the arrival; an assassin's poisoning, with a backup ready, is a cause of the death. Halpern (2015, 8) handles one model of the switch by normality, and Andreas and Günther's deviancy condition does the corresponding work in their theory, as Beckers (2026) shows.

My thesis is that a transferable normality policy faces constraints across cases, even when its scores allow gains in some variables to compensate for losses in others. Componentwise calibration and an additive obstruction make two such constraints precise.

Section 3 derives the componentwise calibration. Section 4 compares switching, short-circuit prevention and preemption by locusts of a normal drought process, then proves the additive obstruction. Section 5 gives a default-preserving isomorphic pair; §6 turns to regularity theories; §7 considers objections.

The claims concern the specified binary models with event variables for the routes to an effect. Richer representations and graded causal judgment remain possible.

Closest work and increment. Halpern and Hitchcock (2015, 19, 34–37) allow exceptions to the default that variables obey their equations. Halpern (2016, Example 3.2.6, 84–85) already shows that excluding a preemption witness can make the preempted throw part of a cause; adding an accuracy variable can sharpen the difficulty. His Theorem 4.4.4(b) (121; proof 133–134) establishes stability under conservative extensions whose added variables respect their equations. The lemma below adapts its elimination-and-minimality reasoning to an individual conforming witness. Neither the problem nor the proof technique is claimed as new.

Halpern also explains why inadequate representations cannot distinguish relevant scenarios (2016, §4.1) and why normality rankings need independent motivation (§4.7). Beckers

(2026) shows that Andreas and Günther's deviancy condition fails for early preemption with a normal backup. These are starting points. The proposed increment is the exact calibration, the three-case comparison under four codings, and the additive obstruction proved below. Hitchcock (2007) supplies the inertial coding and Principle of Sufficient Reason used in §5. The general isomorphism principle is elementary and already familiar.

2. The setting

Models. An acyclic causal model has binary, intervenable variables. A root has no parents among the modeled variables and takes its actual value from the context. A non-root has an equation from its parents. Both are endogenous in the Halpern–Pearl sense; the conformity test here concerns the non-root equations. The actual world is the solution for the actual context; an intervention $[\vec{X} \leftarrow \vec{x}]$ replaces the equations of $\vec{X}$ by constants.

Three definitions. I follow Halpern's (2015, 3–4) statement of the original, updated and modified Halpern–Pearl definitions. $\vec{X} = \vec{x}$ is a cause of φ if both actually hold (AC1), if a minimality condition holds (AC3), and if there is a witness: a setting $\vec{x}'$ and a set $\vec{W}$ of other variables with values $\vec{w}$ such that $[\vec{X} \leftarrow \vec{x}', \vec{W} \leftarrow \vec{w}]\neg\varphi$ (AC2(a)). The original and updated definitions let $\vec{w}$ be any values, subject to sufficiency conditions that differ between them (AC2(b) and AC2(b^u)). The modified definition requires $\vec{w}$ to be the actual values: the witness may only freeze variables at what they actually were (AC2(a^m)). A conjunct of a cause is part of a cause. The witness world is the solution under the witness intervention.

Normality as a filter. Halpern and Hitchcock (2015, 20–21) require a witness world to be at least as normal as the actual world. Their guidelines favor improvements in individual variables' typicality when nothing worsens, but allow special comparisons when some variables improve and others worsen. They also normally treat endogenous variables' agreement with their equations as typical, while permitting exceptions (19–20). I call this second guideline the conformity default. Halpern (2016, §3.2) extends all three definitions this way. His graded account instead retains causes with abnormal witnesses and compares their quality (§3.3); his alternative in §3.5 compares events comprising sets of contexts. The results below concern the world-based filter. They do not directly apply to those alternatives.

Local uniform rankings. Each variable has a default value. In a given world, a non-root variable falls into one of four categories:

- **n**: default, conforming (its value is the default, and its equation gives that value);
- **l**: licensed deviant (its value is deviant, and its equation gives that value);
- **f**: frozen default (its value is the default, but its equation demands the deviant value);
- **u**: unlicensed deviant (its value is deviant, but its equation demands the default).

A root is either at its default or not, and a root's default is more typical than its deviant value. A local uniform ranking is a preorder $\geq$ on $\{\mathbf{n}, \mathbf{l}, \mathbf{f}, \mathbf{u}\}$, the same for every variable. The witness world passes the filter iff each variable's category there is $\geq$ its category in the actual world, a strict componentwise implementation of their first guideline. Their broader framework also permits special cross-variable comparisons; those are outside this class. Three members of the class matter below:

- **Conformity: $\mathbf{n} \equiv \mathbf{l} > \mathbf{f}, \mathbf{u}$.** This makes the conformity default exceptionless.
- **Value-only: $\mathbf{n} \equiv \mathbf{f} > \mathbf{l} \equiv \mathbf{u}$.** Typicality depends on values alone. Halpern and Hitchcock (2015, 35) consider and reject this ranking for their short circuit.
- **License: $\mathbf{l} \equiv \mathbf{n}$, $\mathbf{f} \geq \mathbf{n}$, and $\mathbf{u} \not\geq \mathbf{n}$.** A value is atypical only if it is a deviant value that its own mechanism did not produce.

Coding defaults. A default assignment needs a rule. I consider four uniform rules:

- **Event coding:** occurrences are deviant and absences default.
- **Inertial coding:** a self-sustaining state is default and a change is deviant (Hitchcock 2007, 506–507).
- **Normal-course coding:** a non-root variable's default is the value it takes when every root takes its default value.
- **Normal course given the background:** a non-root variable's default is the value it takes when the candidate cause takes its default value and every other root keeps its actual value.

The last two implement an undisturbed-system conception of defaults reported by Halpern and Hitchcock (2015, 18). Event, inertial and the first normal-course coding assign the same defaults in the four cases of §3; they come apart in §4.

Appendix B distinguishes the initial computational battery from exhaustive checks of the central finite cases.

3. What normality must do

Four familiar cases calibrate the ranking. The theorem is conditional on retaining their stated judgments; it does not establish those judgments independently. Roots are listed with their actual values; every default is 0 under each of the first three coding rules.

Genuine prevention. Roots $P = 1$ (poison in the coffee), $A = 1$ (antidote). $V = P \wedge \neg A$ (the victim dies). The antidote caused the survival.

Chain. Root $C = 1$. $I = C$, $Y = I$. $C = 1$ caused $Y = 1$.

Boulder. Root $F = 1$ (a boulder falls). $B = F$ (the boulder heads for the hiker), $K = F$ (the hiker ducks), $H = B \wedge \neg K$ (the hiker is hit). The ducking caused the hiker not to be hit. (This is Hitchcock's (2001) case, in the form presented by Andreas and Günther (2024, 18).)

Short circuit. Root $A = 1$ (a bodyguard adds an antidote). $P = A$ (a second bodyguard, conspiring, adds poison only if the antidote is in), $V = P \wedge \neg A$. The antidote is not a cause of the survival (Halpern and Hitchcock 2015, 34–35; Halpern 2015, Ex. 3.10).

Theorem 1. Under event, inertial or normal-course coding (with all roots at default), and for each of the three definitions, a local uniform ranking gives the stated verdicts in all four cases iff $\mathbf{l} \equiv \mathbf{n}$, $\mathbf{f} \geq \mathbf{n}$ and $\mathbf{u} \not\geq \mathbf{n}$. That is, iff it is a license ranking.

Proof sketch (Appendix A). Genuine prevention's witness moves V from $\mathbf{n}$ to $\mathbf{l}$, so $\mathbf{l} \geq \mathbf{n}$; the chain's moves I and Y from $\mathbf{l}$ to $\mathbf{n}$, so $\mathbf{n} \geq \mathbf{l}$. The short circuit's only witness keeps the poison in without the antidote (P: $\mathbf{l} \to \mathbf{u}$; V: $\mathbf{n} \to \mathbf{l}$) and must fail, so $\mathbf{u} \not\geq \mathbf{n}$. The ducking's witness either leaves the hiker not ducking as the boulder falls (K: $\mathbf{l} \to \mathbf{f}$) or forces the boulder's course against its equation (B: $\mathbf{l} \to \mathbf{u}$, already excluded), so $\mathbf{f} \geq \mathbf{n}$. A license ranking admits the genuine-prevention, chain and ducking witnesses and excludes the short-circuit witness. □

Of the 75 weak orders, exactly two satisfy the theorem's conditions: $\mathbf{u} < \mathbf{n} = \mathbf{l} = \mathbf{f}$ and $\mathbf{u} < \mathbf{n} = \mathbf{l} < \mathbf{f}$. On a battery of 36 queries in 25 models (Appendix B), both give the standard verdicts under event coding in all three definitions except on Hall's switch and Doorbells (§4) and on one bogus-prevention model, which a neutralization variable corrects, as in Halpern and Hitchcock (2015, 29 fn. 15). Sufficiency is relative to that battery; necessity rests on the four cases alone.

The conformity ranking violates $\mathbf{f} \geq \mathbf{n}$, so it denies, under every definition, that the ducking prevented the hit. The value-only ranking violates $\mathbf{l} \geq \mathbf{n}$ and denies that the antidote

prevented the poisoning. The surviving comparisons systematize exceptions Halpern and Hitchcock explicitly allow: accepting a backup's unlicensed absence in preemption, while excluding poison present against its short-circuit equation (2015, 34–36).

This local simplification has a clear motivation. Halpern's compact-representation rules normally give equal typicality to equation-conforming values and lower typicality to violations, with explicit exceptions and cross-variable comparisons allowed (2016, §5.2.2, 149–150). Theorem 1 asks which distinctions a uniform replacement must retain to fit the four cases. It does not classify all his compact representations.

Halpern and Hitchcock suggest adding intentions variables to remove particular exceptions (2015, 36–37). Under the modified definition, the following witness-level fact limits that strategy. It uses the removal of equation-conforming interventions and AC3 minimality found in Halpern's stability proof (2016, Theorem 4.4.4(b), 133–134). His theorem concerns conservative extensions and normality preservation; the lemma makes no comparison between models and assumes conformity only in the specified witness.

Lemma (frozen witnesses). Under the modified definition, let $\vec{X} = \vec{x}$ be a cause of φ with a witness in which no non-root variable violates its equation. Then every member of $\vec{X}$ is a root, and φ depends counterfactually on $\vec{X} = \vec{x}$.

Proof. Let $\vec{R}$ be the root members of the candidate $\vec{X}$. If the witness violates no non-root equation, it satisfies all the original non-root equations; every root outside $\vec{R}$ retains its actual value, including any frozen root. By acyclicity, uniqueness of solutions therefore makes the witness identical to the solution under $[\vec{R} \leftarrow \vec{r}']$ alone. Since the outcome changes, $\vec{R}$ is nonempty. If $\vec{X}$ contained a non-root member, $\vec{R}$ would be a proper subset satisfying the counterfactual condition with no frozen contingency, contrary to AC3 minimality. Thus every member of the minimal cause is a root, and the outcome changes under its intervention alone. This argument uses minimality; it does not assume that non-root members of a conjunctive candidate have unchanged parents. □

Thus, in an expanded acyclic model where the candidate remains non-but-for or is a non-root minimal cause, every modified-definition witness violates a non-root equation. A changed representation may instead make it a root but-for cause. An added intentions variable cannot rescue an equation-conforming witness while preserving the lemma's assumptions: the modified definition cannot freely reset a contingency variable. The familiar late-preemption consequence, that excluding the frozen backup witness can make Billy's throw part of a cause, is already Halpern's Example 3.2.6 (2016, 84–85). The present point is how the calibration constrains a uniform response: it requires accepting frozen defaults, which creates the switching problem below.

4. Limits of the calibrated local class

Consider first a preemption whose backup is a normal process.

> Locusts. It is the dry season. Roots: $S = 1$ (a locust swarm arrives), $Q = 0$ (rain falls). $D = \neg Q \wedge \neg S$: unwatered and uneaten, the maize dies of drought within the week. $X = S \vee D$: the maize is destroyed. Actually, $D = 0$ and $X = 1$. The swarm destroyed the maize.

Except for the normality of the backup, this is early preemption. Without normality, every account considered here counts the swarm a cause: the three definitions, NESS and CNESS (Beckers 2021), and Andreas and Günther's (2024) preliminary definition. If the backup gets a deviant trigger of its own (a failed irrigation pump, say), all of them count it a cause with normality too (Appendix B).

Now a switch in which neither route is a standing state.

> Doorbells (Gallow 2021, 84–86). Roots: $M = 1$ (Marty presses the button), $T = 1$ (Doc has set the switch to the right). $L = M \wedge \neg T$ (the left bell rings), $R = M \wedge T$ (the right bell rings), $E = L \vee R$ (Einstein barks). Actually, $L = 0$, $R = 1$, $E = 1$. The switch-setting was not a cause of the barking.

Gallow's verdict is that "Doc's flipping the switch to the right was (along with Marty's pressing the button) a cause of the right bell's ringing, but it was not a cause of Einstein's barking" (2021, 85). His model uses a four-valued switch variable; the binary model is my simplification, in the usual format of one event variable per route.

Theorem 2. Fix any local uniform ranking and any of the three definitions. (a) Under event or inertial coding, $T = 1$ is part of a cause of $E = 1$ in Doorbells iff $S = 1$ is part of a cause of $X = 1$ in Locusts. (b) Under either normal-course coding, $S = 1$ is part of a cause of $X = 1$ in Locusts iff $A = 1$ is part of a cause of $V = 0$ in the short circuit. Hence no ranking in the class, under any of the four coding rules, gives all three standard verdicts.

Proof. (a) Under event and inertial coding every non-root variable in both models defaults to 0 (silence and survival are standing states; ringing and dying are changes). The modified witness for the switch-setting is $[T \leftarrow 0, L \leftarrow 0]$, which freezes the left bell silent although the button is pressed. Relative to the actual world, L moves from **n** to **f**, and R and E move from **l** to **n**. The witness for the swarm is $[S \leftarrow 0, D \leftarrow 0]$, which suppresses drought death despite the absent swarm and absent rain. D moves from **n** to **f**, and X from **l** to **n**. Both pass iff $\mathbf{f} \geq \mathbf{n}$ and $\mathbf{n} \geq \mathbf{l}$. The further witnesses that the original and updated definitions allow either make it rain (a root moved to its deviant value) or cancel the press (violating sufficiency); see Appendix A.

(b) Under either normal-course coding the drought death is the default: with the swarm absent and no rain, $D = 1$ and $X = 1$. Actually, D is a licensed deviant (**l**) and X is **n**. In the witness, D becomes **u** (alive, against an equation that now demands death) and X becomes **l**. In the short circuit, under every coding, the antidote's witness moves P from **l** to **u** and V from **n** to **l**. Both pass iff $\mathbf{u} \geq \mathbf{l}$ and $\mathbf{l} \geq \mathbf{n}$.

An exhaustive check confirms both parts for all 355 preorders on the four categories, including partial ones, under all three definitions and with candidates of every possible conjunction size. □

Theorem 2 assumes nothing about which ranking is correct. For the license rule it means: under event or inertial coding, both the swarm and the switch-setting are causes; under normal-course coding the swarm is not. (Under the modified definition the switch-setting's witness also has the profile of Suzy's in late preemption.) An additive score offers another possible exit by allowing compensation; the next result tests it.

Table 1. Verdicts under the modified definition ("C" = cause or part of a cause; the original and updated definitions agree except where noted). Verdicts that depart from the standard one are marked *.

Case (standard verdict)	License, event	License, inertial	License, normal course	Conformity, any coding	Value-only, event
Genuine prevention (C)	C	C	C	C	not*
Boulder: ducking (C)	C	C	C	not*	not*

Case (standard verdict)	License, event	License, inertial	License, normal course	Conformity, any coding	Value-only, event
Short circuit (not)	not	not	not	not	not
Late preemption: Billy (not)	not	not	not	C* (modified only)	not
Locusts (C)	C	C	not*	not*	C
Doorbells (not)	C*	C*	C* / not	not	C*
Hall's LT/RT switch (not)	C*	not	not	not	C*

In the Doorbells row the normal-course codings differ: with all roots at default no bell is due to ring ("C*"); given the actual press, the left bell's ringing is the default ("not").

Halpern (2015, 8; 2016, 90–91) already distinguishes several switch models and explains the effect of normality on them. In the LT/RT route model, excluding the witness where the train fails to take the left track despite a left setting blocks the switch verdict. Under event coding that excludes a frozen default, conflicting with Theorem 1; under inertial coding it can be the license rule's penalty on an unlicensed deviant. Doorbells tests whether that latter policy transfers when both routes are occurrences. The theorem supplies a cross-case constraint within the stated class, rather than discovering that normality can struggle with switches.

Additive compensation

Give a world the sum of its variables' scores. Every root gets $r > 0$ at default and 0 otherwise; every non-root gets a, b, c, d in categories $\mathbf{n}, \mathbf{l}, \mathbf{f}, \mathbf{u}$, respectively. The same five real numbers apply across models. Admit a witness iff its total is at least the actual total. Unlike componentwise comparison, this permits compensation. Additive representations themselves are familiar: Halpern (2016, 144–145) discusses ranking functions with addition as their combination operation. Uniform category scores are the restriction tested here.

Theorem 3. Under event coding and the modified definition, any such score that counts $C = 1$ a cause in Chain and $K = 1$ a cause of survival in Boulder counts $T = 1$ a cause of barking in Doorbells.

Proof. Chain's only singleton witness world is $(C, I, Y) = (0, 0, 0)$, with gain $g_C = r+2a-2b$. Boulder's only singleton witness world is $(F, B, K, H) = (1, 1, 0, 1)$, with gain $g_B = c - a$. Freezing actual values gives no other worlds changing these outcomes. In Doorbells, $[T \leftarrow 0, L \leftarrow 0]$ gives $(M, T, L, R, E) = (1, 0, 0, 0, 0)$, with gain

$$g_T = r + a - 2b + c = g_C + g_B.$$

The two causal judgments require $g_C, g_B \geq 0$, hence $g_T \geq 0$. This valid witness makes the singleton $T = 1$ minimal automatically. □

Thus compensation alone cannot preserve those three judgments. The result does not depend on Locusts or its contested coarse outcome. It still assumes the binary route model, common scores and event coding; variable-specific scores or richer representations can escape it.

The definition restriction matters. With $(r, a, b, c, d) = (2, 0, -1, -5, -3)$, original and updated HP preserve Chain and Boulder while excluding the Doorbells switch. Their additional Boulder witness $(0, 1, 0, 1)$ has gain $r + d - b = 0$; the modified definition cannot change the background root F to 0. The Doorbells witness has gain -1. Enumeration of all candidate subsets confirms that the switch is not even part of a cause under either older definition. The same score accepts genuine prevention and Locusts and excludes short-circuit prevention. The obstruction therefore arises from uniform aggregation together with the modified contingency restriction, not from additivity alone.

Uniformity is stronger than invariance under renaming. Distinct mechanisms can have different statistical reliabilities or normative significance, even when their values share a category. Theorem 3 does not prohibit representing those differences. Its diagnostic is quantitative: if the root-default bonuses for Chain and Doorbells are r_C and r_T, while category scores remain shared, the identity becomes $g_T = g_C + g_B + (r_T - r_C)$. Excluding the switch while retaining both calibration judgments requires $r_C - r_T > g_C + g_B$. Thus even this simple escape has an explicit cost: a sufficiently larger normality bonus for removing the chain's initiating event. Independent evidence might justify that difference. The theorem identifies what a uniform policy cannot supply, and what one non-uniform response must establish.

5. Beyond the class: an isomorph with identical defaults

Halpern and Hitchcock do not commit to the class (2015, 26 fn. 13; 36 fn. 18). The obvious non-local repair looks at a mechanism's inputs: a frozen default is atypical only when some parent of the frozen variable takes a deviant value. This has a pedigree in Hitchcock's (2007, 508) Principle of Sufficient Reason, which says that a variable with parents takes a default value whenever all its parents do; a deviant value needs a deviant source. The drought equation $D = \neg Q \wedge \neg S$ violates that principle under event coding, since a deviant death follows from all-default parents, whereas the left bell's equation does not. This trigger rule penalizes the frozen left bell (its parent, the press, is deviant) but not the frozen maize. With the candidate's own value exempted and the train's left-track travel coded inertially, the trigger rule gives the standard verdict on every case in Table 1 under the original and updated definitions. (Under the modified definition it errs on Billy's throw and on the switch.)

But Hitchcock's principle also tells us how to repair the drought model. A deviant death needs a deviant source, and the obvious one is that the crop is there to die. Make it explicit:

> Seedlings. A field was sown this morning. Roots: $P = 1$ (the field is sown), $S = 1$ (a swarm arrives); it is a dry week, and no rain is possible. $G = P \wedge S$ (the swarm strips the seedlings), $D = P \wedge \neg S$ (the seedlings die of drought within the week), $X = G \vee D$ (the seedlings are destroyed). Actually, $G = 1$, $D = 0$, $X = 1$. The swarm destroyed the seedlings.

The map $P \mapsto M$, $S \mapsto T$, $G \mapsto R$, $D \mapsto L$, $X \mapsto E$ is an isomorphism from Seedlings onto Doorbells. It preserves equations and actual values. It also preserves default assignments under each of the four coding rules:

- Under event and inertial coding, the sowing and the press are actions and so deviant, and every non-root variable defaults to 0.
- Under normal-course coding with all roots at default, nothing is sown and nothing is pressed, so again every non-root variable defaults to 0.
- Given the actual background, the drought death and the left bell's ringing are both defaults, and so are the destruction and the barking.

Proposition 1. Any account whose verdicts depend only on the equations, the actual values and the default assignment (not on what the variables are called) gives the swarm in Seedlings and the switch-setting in Doorbells the same verdict, under each of the four coding rules.

The proof is immediate. Halpern's insufficient-expressiveness diagnosis (2016, 108–111) already explains the general limitation. The pair checks that adding the four specified default assignments does not break this particular isomorphism. Seedlings satisfies Hitchcock's principle, and it is no less a case of a deviant event preempting a normal process. The computed verdicts bear the proposition out: the trigger rule says neither is a cause, the license rule says both are (or, given the actual background, neither), both of Andreas and Günther's deviancy conditions say neither, and NESS and CNESS say both.

An account using only equations, actual values and defaults must break at least one of those represented agreements to separate the pair. Changing the defaults of the shared enabling conditions is one possibility. That does not track the contrast, which survives when the condition is coded alike. Let a postman press the button every noon, so that the press is statistically normal: a switch that diverts a routine signal is still a switch, yet under the trigger rule the switch-setting is now a cause, exactly as the swarm is when the crop is a standing one (Appendix B).

6. Regularity theories

Andreas and Günther's (2024) final theory counts c a cause of e only if, among other conditions, every root proposition left out of a maximal set of facts that does not by itself entail the effect is "more deviant than" its negation (their condition (4), p. 13). In their later definition, as Beckers (2026, 3–4) states it, the condition is restricted to literals whose variables are neither ancestors nor descendants of the candidate (C5). Beckers (2026, 9–11) gives the decisive case against both. In Backup 2 a trainee surgeon operates while a supervisor, as a moral norm requires, stands ready to operate if she does not, and the deviancy definitions "force us to conclude that trainee's action did not cause the outcome" (p. 11). Beckers also proves their 2025 definition equivalent to a 2024 one (Thm. 2.2), so the verdict reaches their current account. The counterexample and the equivalence are Beckers's; I add that no moral norm is needed, and a condition that predicts the failure:

Proposition 2. Let N be a set of actual root literals, none more deviant than its negation, which together with the equations of C's descendants entails E. Then C is not a cause of E under condition (4). If, in addition, no member of N is an ancestor of C, then C is not a cause of E under (C5).

Proof. Both definitions require the equations of all of C's descendants to be among the laws used (their condition (5), Beckers's (C4)). Those laws, together with any set of facts that contains N, entail E, so the maximal non-entailing set of facts must omit some $n \in N$. That n is a root, and it is not more deviant than its negation, so it violates (4). If it is not an ancestor of C, it also violates (C5), because roots are never descendants. □

In Locusts, $N = \{\neg Q\}$: with no rain, the laws of D and X yield $X \leftrightarrow S \vee \neg S$. The absence of rain is statistically normal, violates no norm, and is an absence: normal by each of Andreas and Günther's criteria. One response changes the event/absence classification of the drought, although changing the model or rejecting a verdict also remains possible. Andreas and Günther themselves ask: "Is, for example, the death of a plant an occurring event, or rather the absence of continued living?" (2024, 27). But the same relabelling, applied to the poison-free coffee, would undo their treatment of bogus prevention (pp. 13–14). Proposition

1 adds a representational constraint: when the default assignments correspond under the isomorphism, those resources alone cannot preserve opposite verdicts.

7. Objections and replies

Objection 1: flexibility. Halpern and Hitchcock allow richer rankings, and Halpern supplies graded and event-based alternatives. The local filter is not their whole theory.

Reply. Correct. Halpern's graded account (2016, 85–86) retains causes that a hard filter would exclude. His event-based account (97–99) compares sets of contexts satisfying witness formulas, and his further proposal (100–102) considers changes in normality. None is exhausted by one uniform order on the four local categories. The theorems give no direct impossibility result for them.

Default status and equation conformity offer a compact, transferable specification. Theorems 1 and 2 test componentwise comparison; Theorem 3 also tests compensating additive scores. A richer account must justify its additional distinctions, a requirement already explicit in Halpern (2016, 83, 126–127).

Objection 2: aptness. The binary route models are inapt. Halpern (2015, 7–8; 2016, 38–39) gets the switch right in a three-variable model where the track variable makes no difference to arrival. Seedlings could be modelled the same way, with a single "destroyed by some route" mechanism, or with a finer-grained effect variable.

Reply. Model choice is a serious response. A time-indexed outcome can distinguish destruction on the third day from destruction by the week's end. The swarm can then be a but-for cause of the earlier outcome, while the switch remains irrelevant to whether the bark occurs at its specified time. This uses the importance of temporal representation that Halpern emphasizes (2016, 124–126).

It also limits what the example establishes. If the outcome is only destruction by the week's end, a theorist may deny that the swarm and switch deserve opposite judgments at that grain. The essay therefore offers a conditional diagnostic, not a verdict invariant under every legitimate redescription. Retaining the contrast requires either defending it for the coarse outcome or enriching the representation. Normality cannot recover omitted timing information solely from a set of defaults that agrees across the two models.

Gallow's (2021, 85) multi-valued switch model supplies another representational option; the binary route model here is not his complete model. The question for each option is whether a principled modeling rule selects it before the desired causal verdict is supplied. The displayed pair adds a test case for that familiar modeling question.

Objection 3: the verdicts. Perhaps the swarm is not a cause, since the destruction was coming anyway. Alternatively, with transitivity theorists, perhaps Doc's switch-setting is a cause of the barking.

Reply. The swarm verdict is plausible when the narrative fixes the swarm as the process that actually destroys the maize before drought would have done so. A theorist can dispute that verdict or argue that the equations omit precisely this timing and process information. That is a substantive response, not a contradiction. The results are conditional: if these contrasting judgments are retained while the represented features are held fixed, a default-only filter cannot recover the contrast. An account that accepts causation by the switch avoids that particular demand. Richer process representations remain open.

Objection 4: triviality. Proposition 1 follows from isomorphism, and Theorem 2 rests on one or two comparisons.

Reply. The abstract invariance claim is elementary. Its role is to test a proposed explanatory resource. Defaults can distinguish bogus prevention from overdetermination because the assigned defaults differ. The pair here preserves them under four specified codings, including codings that make a shared enabling condition explicit. The substantive issue is whether the causal contrast survives at that common representational grain. If it does, default status is insufficient; if it does not, the case explains why an appeal to preemption alone did not establish the contrast. Neither answer licenses the unrestricted conclusion that normality never distinguishes isomorphs.

8. Conclusion

Four calibrating judgments force the componentwise license family, which cannot give all the target verdicts. The additive result reaches beyond that aggregation rule: under modified HP, preserving Chain and Boulder forces the Doorbells switch verdict, because its witness gain is exactly the sum of theirs. This obstruction is absent from the original and updated definitions.

These results give explicit constraints within an established discussion of representational limits. A reusable policy must earn its success across cases; merely permitting compensating scores does not always suffice. Additional distinctions, richer models or revised judgments remain available. The results specify where those choices arise, without establishing which response is uniquely correct.

Appendix A. Proofs

Theorem 1, witness enumeration. In genuine prevention, a witness for $A = 1$ sets $A = 0$, possibly with P fixed at 1. The world is $(P, A, V) = (1, 0, 1)$: A is at its default, P is unchanged, and V moves from **n** to **l**. In the chain, the only witness for $C = 1$ is $[C \leftarrow 0]$, under which I and Y move from **l** to **n**; fixing I or Y at 1 restores $Y = 1$.

In the short circuit, $[A \leftarrow 0]$ alone gives $P = 0$ and $V = 0$, so it is not a witness. The only witness is $[A \leftarrow 0, P \leftarrow 1]$, with world $(0, 1, 1)$ for (A, P, V); P moves from **l** to **u** and V from **n** to **l**. It is admitted iff $\mathbf{u} \geq \mathbf{l}$ and $\mathbf{l} \geq \mathbf{n}$.

In the boulder case, $\neg\varphi$ requires $B = 1$ and $K = 0$. With $F = 1$ the world is $(F, B, K, H) = (1, 1, 0, 1)$: K moves from **l** to **f**, B stays **l**, and H moves from **n** to **l**. With F set to 0, B must be set to 1, giving $(0, 1, 0, 1)$: B moves from **l** to **u**, K moves from **l** to **n**, and H from **n** to **l**. Setting H itself violates sufficiency. The four verdicts therefore hold iff (i) $\mathbf{l} \geq \mathbf{n}$, (ii) $\mathbf{n} \geq \mathbf{l}$, (iii) not both $\mathbf{u} \geq \mathbf{l}$ and $\mathbf{l} \geq \mathbf{n}$, and (iv) either $\mathbf{f} \geq \mathbf{l}$ and $\mathbf{l} \geq \mathbf{n}$, or $\mathbf{u} \geq \mathbf{l}$ and $\mathbf{l} \geq \mathbf{n}$. By (i) and (ii), $\mathbf{l} \equiv \mathbf{n}$; then (iii) gives $\mathbf{u} \not\geq \mathbf{n}$, and (iv) reduces to $\mathbf{f} \geq \mathbf{n}$.

The modified definition admits only the first boulder witness, and the conditions are the same. Partial preorders are covered, since only the listed comparisons occur. All four candidates are single conjuncts; conjunctive candidates were checked by machine.

Theorem 2, remaining witnesses. In Locusts, original and updated definitions: a witness setting $Q = 1$ moves a root to its deviant value and fails; setting $D = 0$ gives the modified witness, and $D = 1$ gives $X = 1$. In Doorbells a witness must keep $M = 1$, since with $M = 0$ restoring $T = 1$ does not restore the barking (AC2(b)); then $\neg\varphi$ requires $L = 0$, the modified witness. For conjunctive candidates, exhaustive finite checking covers all 355 category preorders and every nonempty candidate subset, including subsets containing the outcome. It verifies the same equivalences without imposing a size-three cutoff.

Appendix B. The computation

The code (Python, standard library only) implements the three definitions as stated by Halpern (2015, 3–4); Halpern and Hitchcock's filter with componentwise comparison; NESS and CNESS (Beckers 2021); Andreas and Günther's (2024) definitions with condition (4); and the variant with (C5) in Beckers's (2026, 3–4) statement. Without normality it reproduces Halpern's (2015) verdicts in Examples 3.3, 3.4, 3.9 (LT/RT model) and 3.10, and it reproduces Andreas and Günther's (2024, 28–29) table. For each coding rule and each of the 75 weak orders it evaluates 36 queries in 25 models.

The original checks identify the two license weak orders, verify the displayed contrasts and default-preserving isomorphism, and test the regularity-theory corollary on 41 random models. The final targeted check expands the central claims to all 355 reflexive, transitive relations on the four categories. It enumerates every candidate subset, including those containing the outcome, computes every admissible witness under each definition, and applies AC3 after normality filtering. There were no discrepancies in 2,130 calibration comparisons and 3,195 paired-case comparisons. In the cases covered by these counts, inertial coding coincides with event coding. The additive check enumerates singleton witness feature vectors and verifies the exact coefficient identity in Theorem 3. Its boundary counterexample is checked across every candidate subset under all three definitions, using the same HP implementation. The written identity proves the result for all real scores; finite computation supplies a check, not the universal quantifier.

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